UNIVERSITY OF DAR ES SALAAM

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IS335 / CS498: Final Year Project Report –End of Semester One

Project Title:	**Rishfy: A Digital Platform for Ride Sharing**		
Student Name:	Abdallah, Fatma Nassib	2023-04-00047	BSc. Computer Science
	Kahungo, Stella Thomas	2023-04-03737	BSc. Computer Science
	Kaaya, Godbless Gadiel	2023-04-03579	BSc. Computer Science
	Mazwa, Ezekiel Paschal	2023-04-07139	BSc. Computer Science
Supervisor Name:	Dr. Abdulla Ally		
Supervisor Signature:			

ACKNOWLEDGEMENT

In the course of accomplishing this project, we encountered various challenges that tested our commitment, teamwork, and perseverance. We sincerely extend our deepest appreciation to all individuals and institutions whose support made the successful progress of this project possible.

First and foremost, we thank Almighty God for granting us the strength, wisdom, and good health throughout the development of this project.

We would like to express our heartfelt gratitude to our supervisor, Dr. Abdulla Ally, for his continuous guidance, constructive feedback, and unwavering support. His professional insight and encouragement were invaluable in shaping the direction and quality of the Rishfy project.

We are also grateful to the panel of supervisors and the entire Department of Computer Science and Engineering (CSE) at the University of Dar es Salaam(UDSM-COICT) for their academic guidance, motivation, and technical support throughout the semester.

Special appreciation goes to the commuters and private vehicle owners in Dar es Salaam who participated in our survey and informal interviews. Their real-world insights into urban transport challenges were instrumental in validating the need for the Rishfy platform and refining system requirements.

We sincerely thank our lecturers at the University of Dar es Salaam for equipping us with the theoretical and practical knowledge that laid the foundation for this work.

We are deeply thankful to our families for their unconditional love, patience, and encouragement during the course of this project.

Finally, we acknowledge our fellow students, colleagues, and friends for their collaboration, advice, and moral support throughout the development of Rishfy. Your contributions are greatly appreciated.

ABSTRACT

Dar es Salaam is experiencing a growing urban transport crisis characterized by declining public commuter capacity, rising transport costs, safety concerns, and inefficient utilization of private vehicles. Despite the presence of ride-hailing services in the past, regulatory pressures and market shifts have reduced reliable transport options, leaving many daily commuters underserved. Primary survey findings indicate that a large proportion of private vehicle owners frequently travel with empty seats, highlighting a significant coordination gap rather than a shortage of vehicles.

This project presents Rishfy, a mobile-based digital ride-sharing platform designed to connect private vehicle owners with commuters traveling along similar routes in Dar es Salaam. The platform focuses on route-based matching for recurring daily commutes, supported by trust and safety mechanisms such as identity verification, two-way ratings, and real-time GPS tracking. Rishfy is also architected to align with the Land Transport Regulatory Authority (LATRA) D5 ride-sharing framework, positioning it for future regulatory compliance and integration.

The project adopted an Agile development methodology, enabling iterative design, development, and testing. The system employs a three-tier client–server architecture using Flutter for the mobile application, React.js for the administrative interface, Node.js for backend services, and PostgreSQL for data management. Key features include user role management, route posting and matching, seat booking, simulated payment handling, and administrative monitoring tools.

Rishfy addresses critical gaps in existing mobility solutions by prioritizing predictable commute matching, localized trust infrastructure, and regulatory readiness for the Tanzanian context. The project demonstrates the feasibility of leveraging unused private vehicle capacity to improve urban mobility, reduce commuter costs, and enhance transport safety in resource-constrained urban environments.

ACKNOWLEDGEMENT i
ABSTRACT ii
LIST OF ABBREVIATIONS v
CHAPTER ONE: INTRODUCTION 1
1.1 General Introduction 1
1.2 Statement of the Problem 1
1.3 Objectives 2
1.3.1 Main Objective 2
1.3.2 Specific Objectives 2
1.4 Significance of the Project 2
1.5 Project Scope 3
1.5.1 In Scope 3
1.5.2 Out of Scope (Future Enhancements) 3
1.6 Limitations 4
CHAPTER TWO: LITERATURE REVIEW 5
2.1 Introduction 5
2.2 Concept of Shared Mobility 5
2.3 Ride-Sharing and Carpooling Systems 5
2.4 Route-Based Matching Technologies 6
2.5 Trust and Safety in Ride-Sharing Platforms 6
2.6 Review of Existing Related Systems 7
2.7 Research Gaps Identified 7
2.8 Relevance of the Literature to the Current Project 8
2.9 Summary of the Literature Review 8
CHAPTER THREE: METHODOLOGY 9
3.1 Introduction 9
3.2 Data Collection Methods 9
3.3 Problem Validation and Survey Findings 10
3.4 Regulatory Context and LATRA Integration Feasibility 10
3.5 System Modeling and Design 11
3.6 Implementation Approach 11
3.6.1 System Development Methodology 11
3.6.2 System Architecture and Technology Stack 11
3.7 System Testing and Evaluation 12
3.8 Ethical and Security Considerations 12
3.9 Timeline 12
3.10 Chapter Summary 13

CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN 14
4.1 Introduction 14
4.2 SYSTEM ANALYSIS 14
4.2.1 Business Requirements 14
4.2.2 User Requirements 15
4.2.3 Functional Requirements 16
4.2.4 LATRA Regulatory and Compliance Requirements 18
4.2.5 Non-Functional Requirements 21
4.3 SYSTEM DESIGN 23
4.3.1 System Architecture 23
4.3.2 Use Case Modeling 24
4.3.3 Activity Diagrams 25
4.3.4 Sequence Diagrams 28
4.3.5 Entity Relationship Diagram 30
4.4 Chapter Summary 30
REFERENCES 32

LIST OF ABBREVIATIONS

UDSM	University of Dar es Salaam
COICT	College of Information and Communication Technologies
GPS	Global Positioning System
API	Application Programming Interface
JWT	JSON Web Token
LATRA	Land Transport Regulatory Authority
TRA	Tanzania Revenue Authority
TZS	Tanzanian Shilling
PSV	Public Service Vehicle
D5	LATRA Ride-Sharing License Category

CHAPTER ONE: INTRODUCTION

1.1 General Introduction

Dar es Salaam, Tanzania's largest city with a metropolitan population exceeding 6 million, faces a deepening urban transport crisis. The formal public transport system primarily daladalas, is collapsing under growing demand, while the city's population continues to grow at approximately 5% annually. According to the Land Transport Regulatory Authority (LATRA) Annual Statistics Report 2024/2025, commuter bus licenses declined by 12.2% year-over-year (from 11,658 to 10,236), while intercity buses grew by 12.6%, bringing more people into the city. Bodaboda (motorcycle) licenses surged by 15.4%, indicating that commuters are resorting to riskier alternatives.

At the same time, ride-hailing services such as Bolt declined by 10.9% under regulatory pressure, and Uber ceased operations entirely as of January 31, 2026. This has eliminated what was, for many commuters, the most reliable premium transport option. The result is a city where hundreds of thousands of daily commuters struggle to get to work safely, affordably, and on time.

Yet primary survey data collected for this project reveals that 79% of private vehicle owners in Dar es Salaam travel with empty seats 'often' or 'always', and 71% consider this unused capacity a waste of resources. The challenge is not a shortage of vehicles ,it is a coordination failure that prevents drivers and passengers from connecting safely and efficiently.

Rishfy is proposed as a mobile-based digital platform to address this challenge by providing trust infrastructure, route-matching algorithms, and a booking system that makes safe, affordable ride sharing possible at city scale. By leveraging LATRA's newly created D5 ride-sharing license category (established December 2024), Rishfy is designed to operate within Tanzania's evolving regulatory framework from the outset.

1.2 Statement of the Problem

Dar es Salaam's commuting system is experiencing systematic failure driven by three interconnected problems. First, formal public transport infrastructure is collapsing: commuter bus licenses have declined by 12.2% while the population grows at 5% annually, creating an estimated daily deficit of 3 million unmet or delayed passenger trips. Second, alternatives are either unsafe or unaffordable: 83% of surveyed respondents report experiencing high transport

costs or limited route options, and 68% identify physical safety as their primary concern. Bodaboda motorcycles are surging despite well-documented accident risks, while premium ride-hailing has become too expensive or is no longer available. Third – and most critically – 79% of private vehicle owners regularly travel with empty seats, yet no platform exists to safely match these vehicles with commuters traveling the same routes.

There is no centralized, trusted digital platform that efficiently connects vehicle owners with passengers along shared commute routes in a manner that is safe, affordable, regulatory-compliant, and tailored to the Tanzanian context. Rishfy is developed to fill this gap.

1.3 Objectives

1.3.1 Main Objective

To design and develop a digital ride-sharing platform that connects vehicle owners with daily commuters along shared routes, providing a safe, affordable, and LATRA-compliant alternative to collapsing public transport in Dar es Salaam.

1.3.2 Specific Objectives

i. To investigate the urban commute transport problem through user surveys, analysis of existing solutions, examination of LATRA regulatory requirements, and review of relevant literature.

ii. To design a trust-based ride-sharing platform architecture including identity verification mechanisms, route-matching algorithms, user interfaces, and LATRA-compliant technical specifications.

iii. To develop and implement the Rishfy platform with user management, route matching, booking systems, payment integration, and administrative functionalities.

iv. To evaluate the platform's effectiveness through pilot testing, measurement of key performance indicators, validation of trust mechanisms, and assessment of regulatory compliance readiness.

1.4 Significance of the Project

The Rishfy project addresses a critical and immediate urban mobility crisis in Dar es Salaam. Its significance extends across multiple dimensions. For commuters, it provides a safe, reliable, and affordable daily transport option at a time when both daladalas and ride-hailing are failing

simultaneously. For vehicle owners, it unlocks a legitimate income stream from capacity they are already wasting. For the city, it reduces the number of single-occupancy vehicles on high-density corridors such as Mbezi–City Centre and Tegeta–Kariakoo, directly reducing congestion.

From a regulatory perspective, Rishfy is designed to operationalize LATRA's D5 ride-sharing license category, a framework that exists but has had zero licenses issued to date due to the complexity of individual compliance. By acting as a platform operator, Rishfy can lower the barrier to entry for drivers and become Tanzania's first licensed ride-sharing platform. Academically, the project contributes a locally validated, data-driven model for trust-based shared mobility in a Tanzanian urban context.

1.5 Project Scope

1.5.1 In Scope

The project will focus on the following functional areas:

i. User registration and authentication (drivers, passengers, administrators)
ii. User role selection and profile management
iii. Route posting and management by vehicle owners
iv. Listing and booking of available seats
v. Route-matching algorithm based on destination alignment
vi. Real-time GPS tracking during trips
vii. Basic payment handling (simulated/prototype)
viii. Administrative management of the system
ix. Mobile application (Android/iOS via Flutter)

1.5.2 Out of Scope (Future Enhancements)

- In-app messaging between users (Phase 2)
- Luggage and cargo transport features (Phase 2)
- Integration with third-party insurance providers (Phase 2)
- Route optimization across network (Phase 3)
- Web application (Phase 2)
- Full live LATRA system integration API (post-commercialization)

1.6 Limitations

Full LATRA licensing will not be achievable within the academic timeline, as it requires formal business registration and regulatory approvals beyond the project's scope. The platform will instead be developed to be LATRA-compliant in architecture and ready for formal integration upon commercialization. Payment integration will be simulated rather than live. The pilot survey was conducted with 28 respondents, which, while directionally informative, is not a statistically representative sample of Dar es Salaam's commuter population.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to shared mobility systems, ride-sharing and carpooling platforms, route-based matching technologies, and trust mechanisms in peer-to-peer transport services. The purpose is to examine previous studies, existing systems, and technological approaches relevant to the development of Rishfy. By analysing current research and identifying gaps, the chapter provides justification for a commute-focused, trust-enabled ride-sharing platform tailored to the Tanzanian regulatory and social context.

2.2 Concept of Shared Mobility

Shared mobility refers to transportation services that allow users to share vehicles, trips, or transport capacity rather than relying solely on private vehicle ownership. Examples include ride-sharing, carpooling, bike-sharing, and peer-to-peer transport services. Studies consistently demonstrate that shared mobility reduces traffic congestion, lowers per-capita transportation costs, and improves environmental sustainability by reducing single-occupancy vehicle usage (Shaheen, Cohen & Zohdy, 2016).

Shaheen et al. (2016) defined shared mobility as transportation services that enable users to gain short-term access to vehicles, rides, or transport capacity through digital platforms. Their study demonstrated contributions to reduced vehicle ownership, lower transportation costs, and decreased traffic congestion. However, their analysis primarily addressed North American and European contexts, where regulatory frameworks, digital infrastructure, and cultural trust norms differ substantially from Sub-Saharan Africa.

2.3 Ride-Sharing and Carpooling Systems

Ride-sharing platforms such as Ridesharing and carpooling applications such as BlaBlaCar have transformed urban transportation by enabling users to share rides through mobile applications. These systems rely on location tracking, routing algorithms, and real-time matching engines to connect drivers with passengers.

Rayle et al. (2016) evaluated ride-hailing services such as Uber and Lyft, reporting improvements in travel convenience and user accessibility. However, their findings indicate that these platforms primarily focus on immediate, on-demand ride requests and do not effectively integrate pre-planned routes or recurring scheduled travel patterns. This is a critical

gap for daily commuters in cities like Dar es Salaam, where most journeys follow predictable, fixed routes to and from work at consistent times. A platform that optimises for recurring commutes rather than occasional on-demand trips can achieve higher efficiency and better driver-passenger matching.

2.4 Route-Based Matching Technologies

Route-based matching involves aligning the origin, destination, and travel paths of drivers with those of passengers. Previous studies demonstrate that route matching improves efficiency by minimizing detours and reducing travel time. Geospatial technologies such as GPS, Google Maps APIs, and routing algorithms (including shortest-path and proximity-based matching) are commonly employed in these systems.

Agatz et al. (2012) reviewed optimization approaches for dynamic ride-sharing and concluded that effective route-based matching is computationally challenging but essential for maximizing system efficiency. Most existing implementations prioritize real-time ride-hailing over planned route sharing, limiting their ability to support predictable daily commuters. Rishfy's matching algorithm is specifically designed for the planned-route, recurring-trip model that characterises daily urban commuting, where 68% of surveyed vehicle owners report fixed or predictable travel routes.

2.5 Trust and Safety in Ride-Sharing Platforms

Trust is widely identified as the primary barrier to adoption of peer-to-peer transport services, particularly in markets where riders and drivers are strangers. Mechanisms such as identity verification, two-way ratings, profile visibility, and emergency contacts have been shown to significantly increase user willingness to participate in ride-sharing (Teubner & Flath, 2019).

Chinenyeze and Emordi (2022) specifically examined ride-sharing adoption challenges in Sub-Saharan Africa, concluding that trust and safety verification are the primary barriers in African contexts – barriers that global platforms have not adequately addressed. Their study found that cultural norms, lack of digital identity infrastructure, and limited regulatory frameworks make trust-building mechanisms even more critical in these markets than in developed countries. This finding directly shapes Rishfy's design priorities: the platform places identity verification, two-way ratings, and regulatory compliance at the centre of its architecture rather than treating them as secondary features.

2.6 Review of Existing Related Systems

Several existing systems relate to the concepts addressed by Rishfy:

Ride-hailing applications (Uber, Bolt, InDrive) provide on-demand passenger transport but are designed for commercially licensed drivers. Uber has ceased operations in Tanzania as of January 2026, and Bolt has been declining under regulatory pressure. Neither platform supports recurring scheduled commutes or the D5 ride-sharing regulatory framework.

Carpooling platforms (BlaBlaCar) encourage shared rides along pre-planned routes but are largely designed for intercity travel rather than daily urban commutes. They also lack the trust infrastructure and local regulatory alignment required for the Tanzanian market.

Public transport systems (daladalas, DART) offer mass urban mobility but lack flexibility, real-time matching, and the capacity to absorb growing demand as fleet sizes shrink.

No existing platform in Tanzania combines route-based seat matching for daily urban commutes with identity verification, two-way ratings, and alignment with LATRA's D5 regulatory framework.

2.7 Research Gaps Identified

From the reviewed literature, the following gaps are identified that directly justify the Rishfy platform:

- Regulatory Mismatch: Existing platforms operate under commercial taxi or ride-hailing models requiring high licensing costs. Tanzania's LATRA D5 category enables cost-sharing for common routes, yet no platform currently leverages this framework. Zero D5 licenses have been issued to date despite the category's creation in December 2024.
- Commute vs. On-Demand Focus: Global platforms optimize for occasional on-demand trips rather than predictable daily commutes. No system enables recurring bookings that match office workers' fixed schedules and routes.
- Trust Infrastructure: No Tanzania-specific solution combines identity verification, two-way ratings, and local regulatory compliance to address the safety concerns identified by 68% of survey respondents.

- Insufficient Local Context: No existing solution is designed specifically around the transportation realities of Dar es Salaam, including its commuter corridors, Swahili-language context, local payment methods, and LATRA compliance requirements.

2.8 Relevance of the Literature to the Current Project

The reviewed literature provides foundational concepts and technological approaches that directly inform Rishfy's design. Shared mobility theory validates the market opportunity; ride-sharing research highlights the importance of planned-route matching over real-time demand; trust literature, particularly Chinenyeze and Emordi (2022), establishes the primacy of verification and ratings in African contexts; and regulatory analysis confirms that Tanzania's D5 framework creates a first-mover opportunity. Together, these sources guide the platform's architecture toward a commute-focused, trust-first, locally compliant solution.

2.9 Summary of the Literature Review

This chapter examined existing studies, technologies, and systems related to shared mobility, ride-sharing, route matching, and trust mechanisms in peer-to-peer transport. The review confirmed that while multiple shared mobility solutions exist globally, none adequately addresses the specific combination of daily urban commute matching, trust infrastructure, private vehicle integration, and LATRA regulatory compliance required in Tanzania. These findings justify the development of Rishfy as a novel, locally validated platform.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter describes the methods and procedures used in carrying out the Rishfy project. It explains how data was collected, how system requirements were identified and analysed, and how the project objectives will be implemented through systematic system design, development, and testing. The methodology provides a clear account of the steps followed from problem identification to the development of the final digital platform.

3.2 Data Collection Methods

Both primary and secondary data sources were used to gather information relevant to the development of Rishfy.

3.2.1 Primary Data Collection

Primary data was collected to understand real-world transportation challenges, user needs, and willingness to participate in ride sharing. The following methods were employed:

Survey: A structured questionnaire was administered to 28 respondents in Dar es Salaam comprising both vehicle owners and daily commuters. The survey collected quantitative data on commuting patterns, transport costs, safety perceptions, regulatory awareness, and willingness to use a ride-sharing platform.

Observation: Daily commuting patterns along key corridors (Mbezi–City Centre, Tegeta–Kariakoo) were observed to assess vehicle utilisation, boarding behaviour at bus stops, and traffic distribution.

Informal Interviews: Conversations with private car owners, passengers, and transport-sector stakeholders were conducted to understand barriers to participation, particularly around regulatory compliance and safety concerns.

Primary data helped identify practical system requirements and validated both the demand-side (commuter pain points) and supply-side (driver willingness) of the platform.

3.2.2 Secondary Data Collection

Secondary data was collected through a review of existing literature and official sources, including:

- LATRA Annual Statistics Report 2024/2025 for transport fleet data and regulatory context
- LATRA Private Hire Services Regulations (2020) and 2024 Amendments for compliance requirements
- Academic journals on shared mobility, ride-sharing adoption, and trust mechanisms in Sub-Saharan Africa
- Technical documentation of existing platforms and route-matching technologies

Secondary data informed the understanding of the regulatory environment, existing solutions, and technological best practices.

3.3 Problem Validation and Survey Findings

The primary survey of 28 respondents produced key validation data. On the demand side, 83% reported experiencing high transport costs or limited route options, 68% identified safety as their primary transport concern, and 86% said they would use a platform matching drivers and passengers along the same route. On the supply side, 79% of vehicle owners reported travelling with empty seats often or always, 71% felt this was a waste of resources, 79% said extra income would motivate them to offer transport services, and 68% reported having fixed or predictable routes suitable for recurring bookings.

Key barriers identified by respondents include the cost and complexity of TRA vehicle change-of-use and LATRA licensing requirements (mentioned by 15 respondents), unclear regulations (8 mentions), and safety concerns around inviting strangers into private vehicles. These findings directly shaped Rishfy's regulatory strategy and trust architecture.

3.4 Regulatory Context and LATRA Integration Feasibility

LATRA created a Ride-Sharing category (D5) in December 2024, explicitly designed for cost-sharing on common destination routes. However, zero D5 licenses have been issued to date, indicating that high barriers prevent individuals from navigating compliance independently. Rishfy's regulatory strategy is to apply for a D5 operator license, provide compliance support to drivers, and become Tanzania's first licensed ride-sharing platform operator.

Full LATRA licensing will not be achievable within the academic timeline due to formal business registration requirements. However, within the project timeline, the technical and operational integration groundwork will be completed: building app functionality meeting all

LATRA operational criteria, developing API architecture aligned with LATRA specifications, implementing OAuth 2.0 authentication, structuring all required trip data in LATRA-compliant JSON format, using mock LATRA endpoints to simulate integration, and preparing complete compliance documentation.

3.5 System Modeling and Design

The analysed requirements were translated into system models using UML diagrams, including Use Case Diagrams to represent interactions between users and the system, Class Diagrams to define system structure and data relationships, and Sequence Diagrams to illustrate the flow of interactions during key processes such as route matching and booking. User interface designs are being created in Figma to visualise system workflows prior to implementation.

3.6 Implementation Approach

3.6.1 System Development Methodology

The project adopts an Agile development methodology, allowing incremental development and continuous refinement. Development is carried out in iterations consisting of feature development, testing, and review. This approach ensures flexibility, early identification of issues, and alignment with evolving requirements.

3.6.2 System Architecture and Technology Stack

Rishfy is implemented using a client–server architecture. The technology stack comprises:

Layer	Technology
Mobile Frontend	Dart / Flutter (iOS & Android)
Web/Admin Frontend	JavaScript, React.js
Backend	Node.js with RESTful APIs and managed edge functions
Database	PostgreSQL (relational data storage)
Authentication	JSON Web Tokens (JWT) / OAuth 2.0
Maps & Routing	Google Maps API / GPS
UI/UX Design	Figma

System Modeling	Draw.io (UML diagrams)
Version Control	GitHub

3.7 System Testing and Evaluation

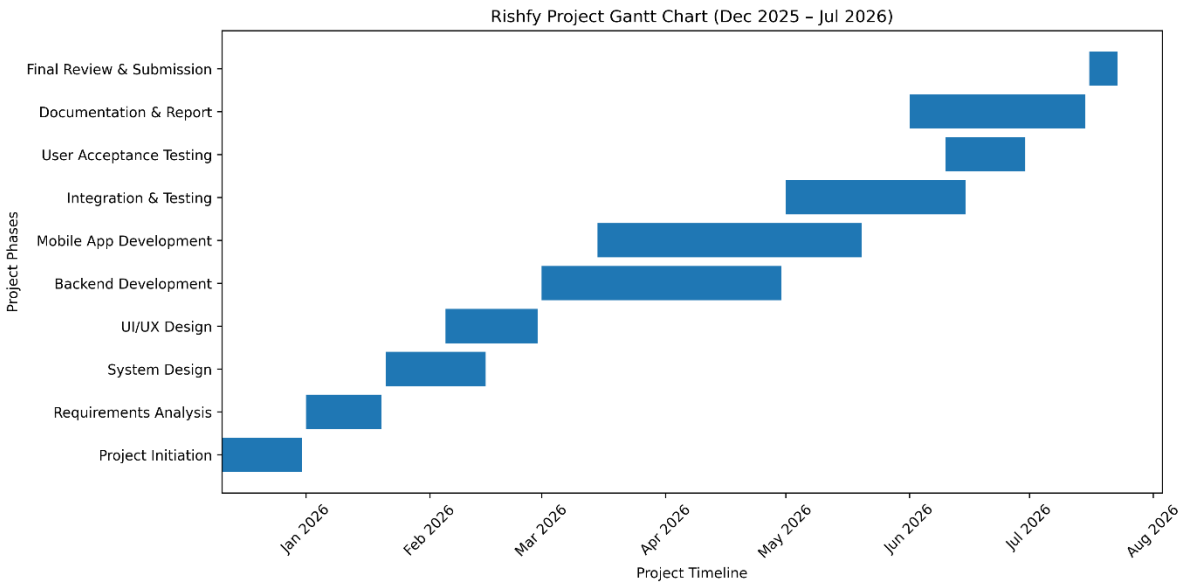
Testing will ensure that the system meets its functional and non-functional requirements. Unit testing will verify individual modules in isolation. Integration testing will assess interactions between system components (e.g., matching algorithm, booking engine, payment handler). User Acceptance Testing (UAT) will involve sample commuters and vehicle owners evaluating usability, correctness, and trust mechanism effectiveness. System evaluation will measure key performance indicators including route-matching accuracy, booking success rate, and user satisfaction scores.

3.8 Ethical and Security Considerations

The project ensures secure handling of all user data, including personal profiles and real-time location information. Access is controlled through JWT-based authentication. User consent is obtained for data collection and sharing. The platform implements two-way identity verification to protect both drivers and passengers. Location data is transmitted and stored securely and used exclusively for trip management purposes.

3.9 Timeline

Phase	Start	End	Duration
Initiation	11 Dec 2025	31 Dec 2025	3 weeks
Requirements	01 Jan 2026	20 Jan 2026	3 weeks
Design	21 Jan 2026	28 Feb 2026	6 weeks
Development	01 Mar 2026	20 May 2026	11 weeks
Testing	01 May 2026	30 Jun 2026	8 weeks
Documentation	01 Jun 2026	15 Jul 2026	6 weeks
Final Submission	16 Jul 2026	23 Jul 2026	1 week



3.10 Chapter Summary

This chapter outlined how the Rishfy project will be carried out, from primary and secondary data collection through problem validation, requirement analysis, system modelling, architecture design, implementation methodology, and testing procedures. The methodology is designed to ensure the platform effectively addresses the identified transport problem, meets LATRA compliance readiness requirements, and delivers a trustworthy and usable experience for Dar es Salaam commuters.

CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents a comprehensive analysis and design of the Rishfy platform. It begins with system analysis, detailing business, user, functional, and non-functional requirements derived from the problem statement, survey findings, and stakeholder consultations. The chapter then proceeds to system design, presenting architectural models, data flow diagrams, activity diagrams, sequence diagrams, entity-relationship diagrams, and interface designs. These models collectively provide a blueprint for implementing a secure, scalable, and user-friendly ride-sharing platform aligned with LATRA regulatory requirements and Dar es Salaam's urban transport needs.

4.2 SYSTEM ANALYSIS

4.2.1 Business Requirements

Business requirements define the high-level objectives that the Rishfy platform must fulfill to address the identified transport crisis and deliver value to stakeholders. These requirements are derived from the problem statement, LATRA regulatory framework, and market analysis.

BR1: Market Need Fulfillment

The platform must address the structural transport deficit in Dar es Salaam by connecting the 79% of private vehicle owners traveling with empty seats to the 83% of commuters experiencing high costs or limited route options. This creates a viable middle option between unreliable daladalas and expensive ride-hailing.

BR2: Regulatory Compliance Readiness

The system must be designed to operate within Tanzania's legal framework, specifically LATRA's D5 ride-sharing category created in December 2024. The platform architecture must support eventual integration with LATRA's systems through OAuth 2.0 authentication, LATRA-compliant JSON trip data structures, and documented API readiness.

BR3: Trust and Safety Infrastructure

Given that 68% of surveyed users identify physical safety as their primary transport concern, the platform must implement robust identity verification, two-way rating systems, real-time trip tracking, and emergency features to enable strangers to safely share vehicles.

BR4: Revenue Model Viability

The platform must support a sustainable revenue model through commission-based transactions, enabling monetization while keeping passenger costs affordable and driver earnings competitive. The system must track all transactions for revenue reporting and support multiple payment methods aligned with Tanzanian practices.

BR5: Scalability and Growth

The platform architecture must support scaling from pilot deployment to city-wide adoption, handling thousands of concurrent users, routes, and bookings. The design must accommodate future expansion to other Tanzanian cities and additional features such as luggage transport and intercity routes.

4.2.2 User Requirements

User requirements specify what each user category needs to accomplish using the system. These are derived from primary survey data (N=28), informal interviews, and observation of commuting patterns.

Driver (Vehicle Owner) Requirements:

- Register and manage vehicle details including registration number, photos, and LATRA compliance documents
- Create recurring or one-time routes along their daily commute paths
- Specify departure times, available seats, and price per seat
- Receive and review booking requests from passengers
- Accept or reject bookings based on passenger profiles and ratings
- Receive payment credits after completed trips
- View booking history and earnings summaries
- Rate and review passengers after trips
- Receive notifications for new bookings and trip reminders

Passenger (Commuter) Requirements:

- Search for available routes by entering destination and travel date
- View matched routes with driver profiles, vehicle details, ratings, and pricing
- Book seats on preferred routes
- Make secure payments via mobile money or other supported methods
- Receive trip confirmation and driver contact details
- Track trip in real-time using GPS during journey
- Rate and review drivers after completed trips
- View booking history and saved routes
- Receive notifications for booking confirmations and trip updates

Administrator Requirements:

- Manage user accounts (view, suspend, verify, delete)
- Verify driver and vehicle documentation
- Monitor active trips and booking transactions
- Generate compliance reports for LATRA submission
- Handle dispute resolution between drivers and passengers
- View platform analytics including user growth, trip volumes, and revenue
- Configure system parameters and pricing rules

4.2.3 Functional Requirements

Functional requirements define specific behaviors and capabilities the system must implement. These are organized by module and assigned unique identifiers for traceability.

User Management Module:

ID	Requirement Description
FR-UM-01	The system shall allow users to register using email or phone number
FR-UM-02	The system shall send OTP verification codes for phone number validation
FR-UM-03	The system shall authenticate users using email/phone and password
FR-UM-04	The system shall generate and manage JWT tokens for session management
FR-UM-05	The system shall allow users to select role (Driver, Passenger, or both)

FR-UM-06	The system shall enable users to upload profile pictures and update bio
FR-UM-07	The system shall display user average ratings and total trip counts
FR-UM-08	The system shall allow administrators to verify, suspend, or delete accounts

Vehicle and Route Management Module:

ID	Requirement Description
FR-VR-01	The system shall allow drivers to register vehicles with details (make, model, year, color, registration)
FR-VR-02	The system shall enable drivers to upload vehicle photos and documents
FR-VR-03	The system shall allow drivers to create routes specifying origin, destination, departure time
FR-VR-04	The system shall support recurring routes with daily, weekly, or custom patterns
FR-VR-05	The system shall enable drivers to set available seats and price per seat
FR-VR-06	The system shall validate route data (coordinates, dates, pricing) before saving
FR-VR-07	The system shall allow drivers to edit or deactivate posted routes
FR-VR-08	The system shall display route status (Active, Full, Completed, Cancelled)

Route Matching and Booking Module:

ID	Requirement Description
FR-RM-01	The system shall match passenger destinations with available driver routes within 2km proximity
FR-RM-02	The system shall rank matched routes by departure time, price, and driver rating
FR-RM-03	The system shall display route details including driver profile, vehicle info, and passenger reviews
FR-BK-01	The system shall allow passengers to book available seats on matched routes
FR-BK-02	The system shall prevent double-booking by checking real-time seat availability
FR-BK-03	The system shall notify drivers of new booking requests
FR-BK-04	The system shall allow drivers to accept or reject booking requests within 2 hours
FR-BK-05	The system shall automatically cancel bookings if driver does not respond within timeout period
FR-BK-06	The system shall enable passengers to cancel bookings up to 2 hours before departure

Payment and Trip Tracking Module:

ID	Requirement Description
FR-PY-01	The system shall process payments via mobile money (simulated in prototype)
FR-PY-02	The system shall generate unique transaction references for all payments
FR-PY-03	The system shall update booking status to 'Confirmed' upon successful payment

FR-PY-04	The system shall credit driver accounts after trip completion
FR-TR-01	The system shall enable real-time GPS tracking during active trips
FR-TR-02	The system shall log location coordinates at 30-second intervals
FR-TR-03	The system shall share live driver location with booked passengers
FR-TR-04	The system shall allow drivers to mark trips as 'Started' and 'Completed'
FR-TR-05	The system shall calculate trip duration and distance upon completion

Rating and Notification Module:

ID	Requirement Description
FR-RT-01	The system shall enable two-way ratings (1-5 stars) after completed trips
FR-RT-02	The system shall allow users to write optional review text with ratings
FR-RT-03	The system shall calculate and display user average ratings
FR-RT-04	The system shall update user ratings in real-time after new reviews
FR-NT-01	The system shall send push notifications for booking requests, confirmations, and trip updates
FR-NT-02	The system shall send SMS notifications for critical events (booking confirmation, trip start)
FR-NT-03	The system shall mark notifications as read/unread and maintain notification history

4.2.4 LATRA Regulatory and Compliance Requirements

In addition to user-facing requirements, the Rishfy platform must satisfy regulatory requirements established by the Land Transport Regulatory Authority (LATRA) to operate legally in Tanzania. While LATRA's published guidelines focus on ride-hailing services, the D5 ride-sharing category introduced in December 2024 follows a similar compliance framework.

1. Operational Requirements

LATRA mandates that the ride-sharing application must implement specific operational features to ensure passenger safety, fare transparency, and service accountability. These operational requirements directly inform Rishfy's functional specifications:

- LATRA-OR-01: Customer Registration and Verification

The system must require passenger registration with personal details collection and OTP (One-Time Password) verification via SMS to enhance security and prevent fraudulent accounts. Rishfy implements this through FR-UM-01 (user registration with phone/email) and FR-UM-02 (OTP verification codes).

- LATRA-OR-02: Vehicle Registration Details

The platform must capture and display complete vehicle information including registration number, make, model, color, and LATRA license status. This is addressed through FR-VR-01 (vehicle registration with details) and the VEHICLE entity in the database schema which includes fields for latra_verified and latra_license_number.

- LATRA-OR-03: Driver Details Display

Driver name and photo must be visible to passengers before booking confirmation. Rishfy's USER entity stores profile_picture and biographical information, displayed in route listings as specified in FR-RM-03 (display route details including driver profile).

- LATRA-OR-04: Accurate Location Setting

The system must enable passengers to set accurate pickup and destination coordinates using GPS or map selection. This requirement is fulfilled through FR-VR-03 (routes specify origin and destination coordinates) and integration with Google Maps API for geocoding.

- LATRA-OR-05: Real-Time Arrival Estimate

The application must provide time estimates for driver arrival at pickup points after trip acceptance. While Rishfy focuses on pre-scheduled commute routes rather than real-time hailing, the system satisfies this through departure time display and notification when drivers are approaching scheduled pickup times.

- LATRA-OR-06: Vehicle Movement Tracking

The platform must show vehicle movement to pickup and destination points with route visualization. Rishfy implements this via FR-TR-01 (real-time GPS tracking during trips), FR-TR-02 (location logging at 30-second intervals), and FR-TR-03 (sharing live location with passengers).

- LATRA-OR-07: Trip Cancellation Policy

The system must not charge passengers when trips are cancelled before the driver arrives at the pickup point. This is addressed through FR-BK-06 (passengers can cancel up to 2 hours before departure with refund) and the PAYMENT entity status field which tracks payment states.

- LATRA-OR-08: Emergency Contact Feature

The application must provide emergency contact functionality for passenger safety. While not explicitly detailed in current functional requirements, this will be implemented as a safety button triggering notifications to pre-designated emergency contacts and administrators.

- LATRA-OR-09: Location-Based Fare Calculation

Charges must be based on actual trip endpoints rather than pre-booking estimates. Rishfy's pre-agreed pricing model (drivers set price per seat on fixed routes) satisfies this through transparent fare display before booking confirmation, with FR-TR-05 (calculate distance upon completion) enabling post-trip verification.

- LATRA-OR-10: Digital Receipt Generation

The system must generate receipts after trip completion showing starting point, ending point, fare, and time. This is fulfilled through the PAYMENT entity which stores transaction_reference, amount, and payment_date, combined with TRIP entity data (start/end coordinates and times).

- LATRA-OR-11: Trip Status Notifications

Passengers must receive notifications when drivers arrive, trips start, and trips end. Rishfy implements this through FR-NT-01 (push notifications for booking events) and FR-NT-02 (SMS notifications for critical events including trip start).

2. Technical Integration Requirements

LATRA requires platform operators to develop an API that integrates with LATRA's monitoring systems, enabling regulatory oversight of trip data and driver earnings. The API must accept date range queries and return completed trip data in standardized JSON format with OAuth 2.0 authentication. The following technical specifications are mandated:

No.	LATRA Data Field	Rishfy Implementation Mapping
i.	**Trip ID**	TRIP.trip_id (primary key, integer, auto-increment)
ii.	**Origin Coordinates**	ROUTE.origin_lat, ROUTE.origin_lng (decimal, comma-separated in JSON export)
iii.	**End Coordinates**	ROUTE.destination_lat, ROUTE.destination_lng (decimal, comma-separated)

iv.	**Start Time**	TRIP.start_time (timestamp, formatted without 'T' or decimal seconds)
v.	**End Time**	TRIP.end_time (timestamp, formatted per LATRA specification)
vi.	**Total Fare Amount**	PAYMENT.amount (decimal, Tanzanian Shillings)
vii.	**Trip Distance**	TRIP.distance (integer, meters, calculated via Google Maps Distance Matrix API)
viii.	**Rating**	RATING.rating_value (integer 1-5, average if multiple passengers)
ix.	**Driver's Earning**	PAYMENT.amount minus platform commission (calculated field)
x.	**Driver License Number**	USER.license_number (to be added to USER entity for drivers)
xi.	**Vehicle Registration**	VEHICLE.registration_number (string, unique)

LATRA API Integration Architecture:

Rishfy's system architecture includes a dedicated Reporting Service (as shown in Figure 4.2: System Architecture) responsible for LATRA compliance. This service will implement the following integration pattern:

- Endpoint: GET /api/v1/latra/trips
- Query Parameters: start_date (ISO 8601 format), end_date (ISO 8601 format)
- Authentication: OAuth 2.0 with LATRA-issued client credentials
- Response Format: JSON array of trip objects containing all 11 required fields
- Data Transformation: Timestamps formatted without 'T' separator; coordinates comma-separated; distance in meters (not kilometers)
- Rate Limiting: Configurable to LATRA specifications (anticipated: 100 requests per hour)

Vehicle Verification API:

LATRA provides a Vehicle Verification API to operators for validating LATRA-licensed vehicles before onboarding. Rishfy will integrate this during driver vehicle registration (FR-VR-01) to verify registration numbers against LATRA's database and update the VEHICLE.latra_verified boolean field. This ensures only compliant vehicles are permitted to post routes.

4.2.5 Non-Functional Requirements

Non-functional requirements specify quality attributes and constraints that the system must satisfy. These ensure the platform is secure, reliable, scalable, and compliant

ID	Requirement	Description	Target
NFR-01	**Performance**	The system must be optimized for responsiveness on mobile networks (3G/4G) common in the operating region. The application should minimize data usage and provide immediate visual feedback (loaders, skeletons.	• APIs respond within ~2s on average • App loads on 3G within 3-5 seconds • Visual indicators appear immediately (0.1s) after user input. • Low-bandwidth mode for map rendering.
NFR-02	**Security**	Security must be built-in by default, ensuring all communications between the app, web portal, and backend are encrypted. Access controls must prevent unauthorized data access, and sensitive securely.	• 100% of traffic over HTTPS (TLS 1.2+) • Zero plaintext passwords stored (Bcrypt/Argon2) • API Rate limiting (e.g., 60 req/min) • Session timeouts for inactive admin users.
NFR-03	**Reliability & Availability**	The system must prioritize uptime during core business hours. It must include a strategy for "graceful degradation," meaning if a third-party service fails, the app allows the user to continue rather than crashing.	• 99.5% availability • Automated database backups every 24 hours • "Retry" mechanisms implemented for failed network requests • User notified clearly if external systems are down.
NFR-04	**Usability**	The interface design must be intuitive for users with varying levels of digital literacy. Error handling must be friendly and actionable, guiding the user on how to fix the problem rather than showing technical error codes.	• Complete booking flow possible in < 5 taps. • Support for Android 8+ and iOS 13+. • Error messages explain *why* an action failed (e.g., "Insufficient funds" vs "Error 500").
NFR-05	**Scalability**	The backend architecture should be stateless and modular, separating the API layer from the database. This ensures the infrastructure can be vertically scaled or migrated to larger cloud instances seamlessly.	• Architecture supports separating App and DB servers • Stateless authentication (JWT) to allow future load balancing. • Database schema allows for future indexing/optimization.
NFR-06	**Compliance**	Comprehensive audit logs must be maintained for all financial transactions and critical booking changes to ensure accountability and facilitate dispute resolution.	• Trip data structure compatible with LATRA requirements • Immutability for all completed payment records. • Privacy policy accessible within the app.

4.3 SYSTEM DESIGN

4.3.1 System Architecture

Rishfy employs a three-tier client-server architecture comprising the presentation layer (mobile and web clients), application layer (business logic and services), and data layer (database and storage). The architecture is designed for modularity, scalability, and maintainability.

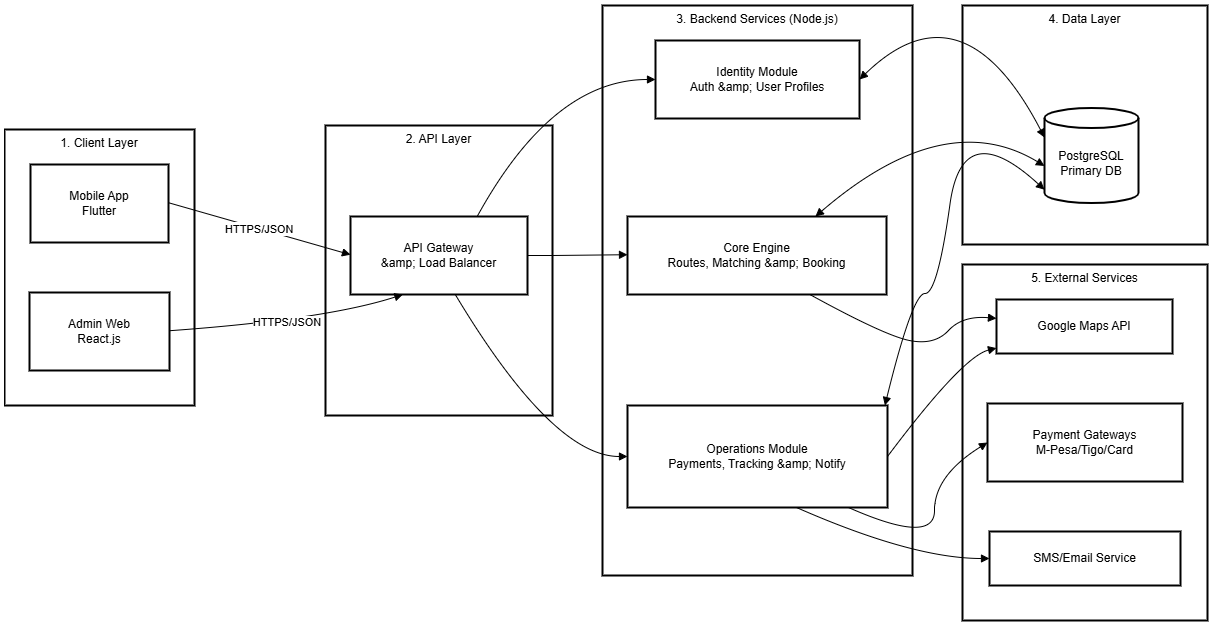


Figure 4.1: System Architecture Overview

The Client Layer consists of a Flutter-based mobile application for drivers and passengers, and a React.js web dashboard for administrators. Both communicate with the backend via RESTful APIs over HTTPS.

The Application Layer implements core business logic through microservices: User Service handles registration and profiles; Route Service manages route CRUD operations; Matching Engine performs destination-based route matching using geospatial algorithms; Booking Service coordinates seat allocation and reservation logic; Trip Service manages real-time tracking; Payment Service handles transaction processing; Notification Service sends push notifications, SMS, and emails; Rating Service manages reviews and reputation scoring; and Reporting Service generates analytics and compliance reports for administrators and LATRA.

The Data Layer uses PostgreSQL for primary transactional data with ACID guarantees, Redis for session caching and real-time data, and cloud file storage for user photos and documents.

External integrations include Google Maps API for geocoding and routing, payment gateways for mobile money transactions, and SMS/email services for notifications.

4.3.2 Use Case Modeling

The use case diagram identifies the key actors (Driver, Passenger, Administrator, LATRA, Payment Gateway) and their interactions with the Rishfy platform. Use cases are grouped into authentication, driver functions, passenger functions, shared functions, and administrative functions.

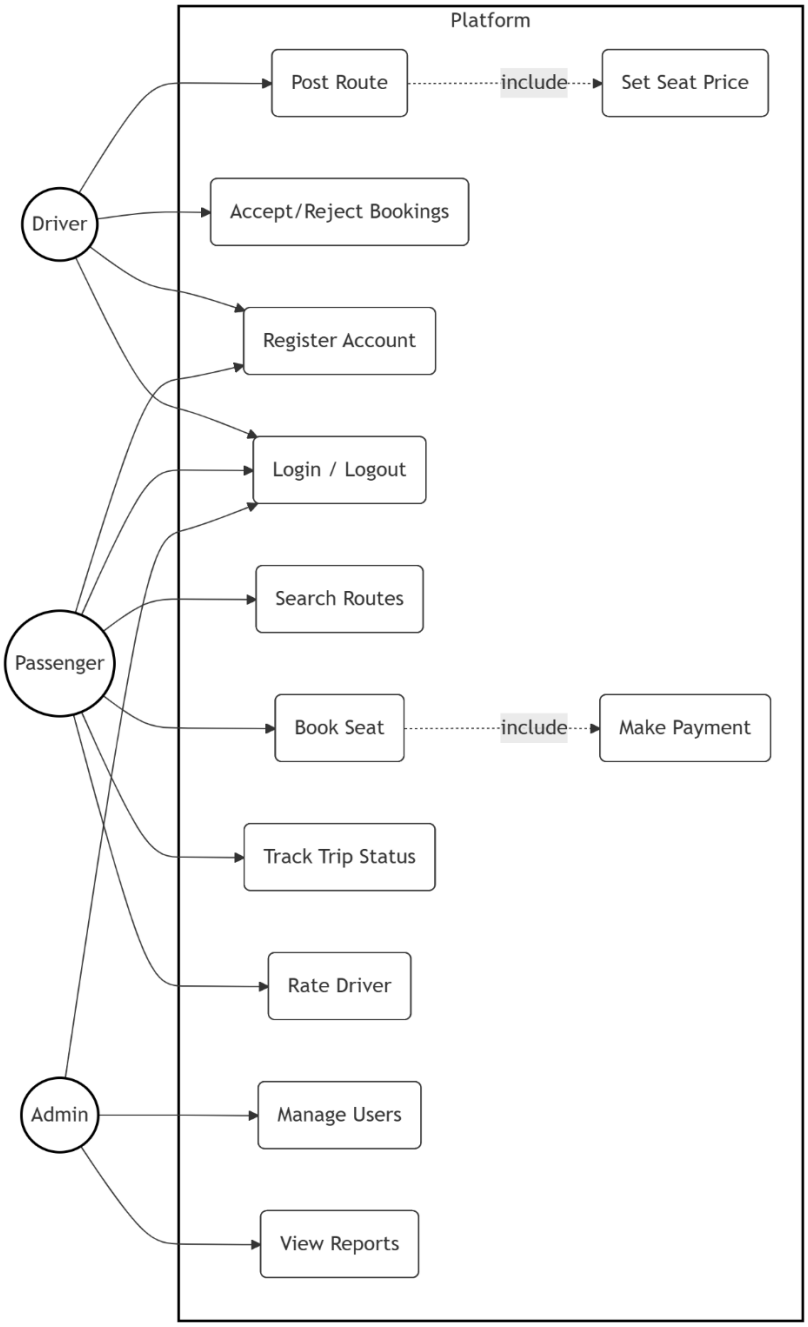


Figure 4.2: System Use Case Diagram

Driver use cases include vehicle registration, route posting and management, booking acceptance/rejection, trip initiation, and trip completion. Passenger use cases include route searching, route detail viewing, seat booking, payment processing, real-time tracking, and booking cancellation. Shared use cases available to both roles include rating and reviewing, viewing trip history, receiving notifications, and reporting issues. Administrator use cases include user management, driver verification, trip monitoring, report generation, dispute handling, and system configuration. LATRA receives compliance reports, while the Payment Gateway processes financial transactions.

4.3.3 Activity Diagrams

Activity diagrams model the workflows for critical user processes. Two key activities are driver route posting and passenger booking.

Driver Route Posting Activity:

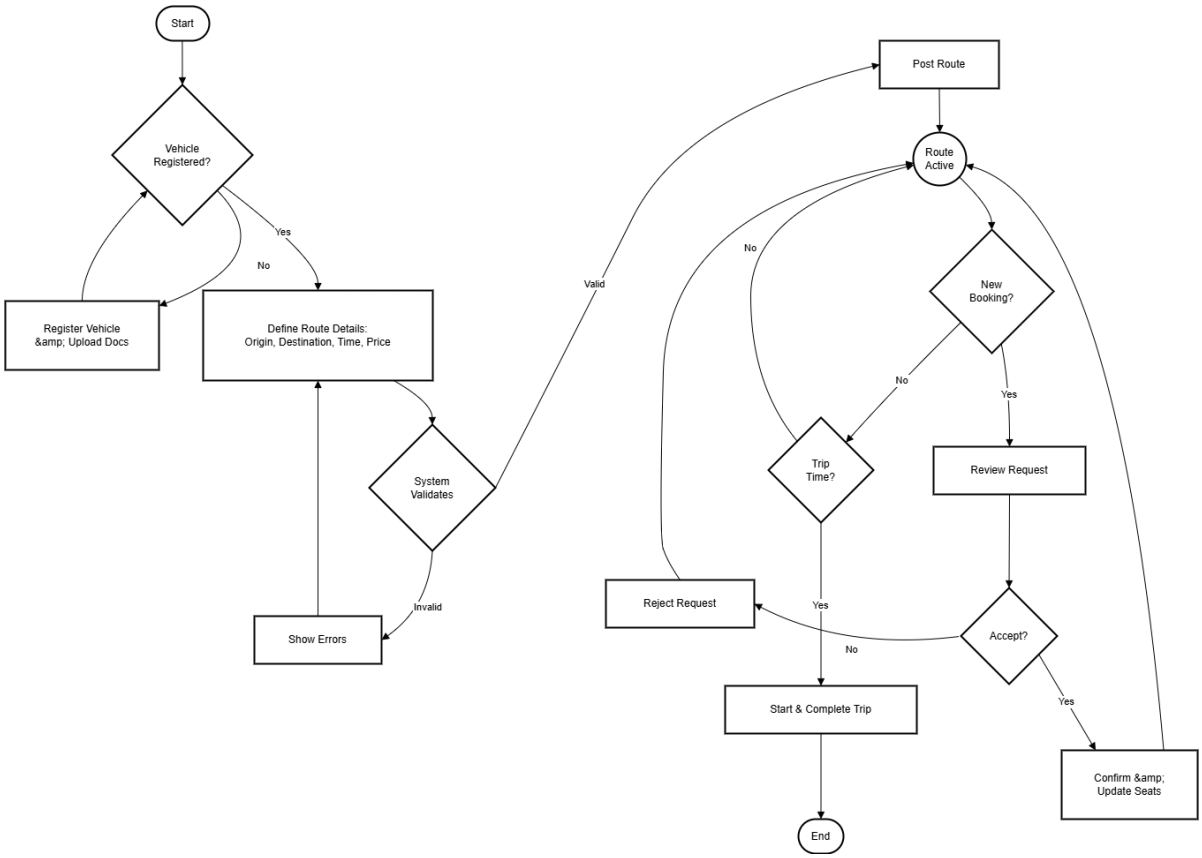


Figure 4.5: Driver Route Posting Activity Flow

The driver opens the app and logs in. If no vehicle is registered, the driver adds vehicle details and uploads documents. Once authenticated with a registered vehicle, the driver selects 'Create

New Route' and enters origin location, pickup points, destination location, and dropoff points. The driver then selects departure date and time, and indicates whether the route is recurring (with a recurrence pattern) or one-time. The driver specifies available seats and price per seat, then reviews the complete route details. Upon confirmation, the system validates the route data. If validation fails, errors are shown and the driver corrects them. If valid, the route is saved to the database and potential passengers are notified. The route becomes active and the driver can manage it by editing, viewing bookings, accepting or rejecting booking requests, or deactivating the route.

Passenger Booking Activity:

The passenger opens the app and logs in. After authentication, the passenger enters destination and travel details. The system matches available routes using the matching engine. If no routes are found, the passenger is notified and can modify search criteria. If routes are found, they are displayed with driver profiles, vehicle details, and ratings. The passenger selects a preferred route and views detailed driver and vehicle information. Upon confirming the booking, the system checks seat availability in real-time. If seats are already booked, an error is shown. If seats are available, a booking record is created and the driver is notified. The driver can accept or reject the request. If rejected, the booking is cancelled and the passenger returns to route selection. If accepted, the passenger proceeds to payment. Successful payment confirms the trip and sends confirmation details to both parties. On the trip day, real-time GPS tracking is enabled. Upon trip completion, the passenger rates and reviews the driver.

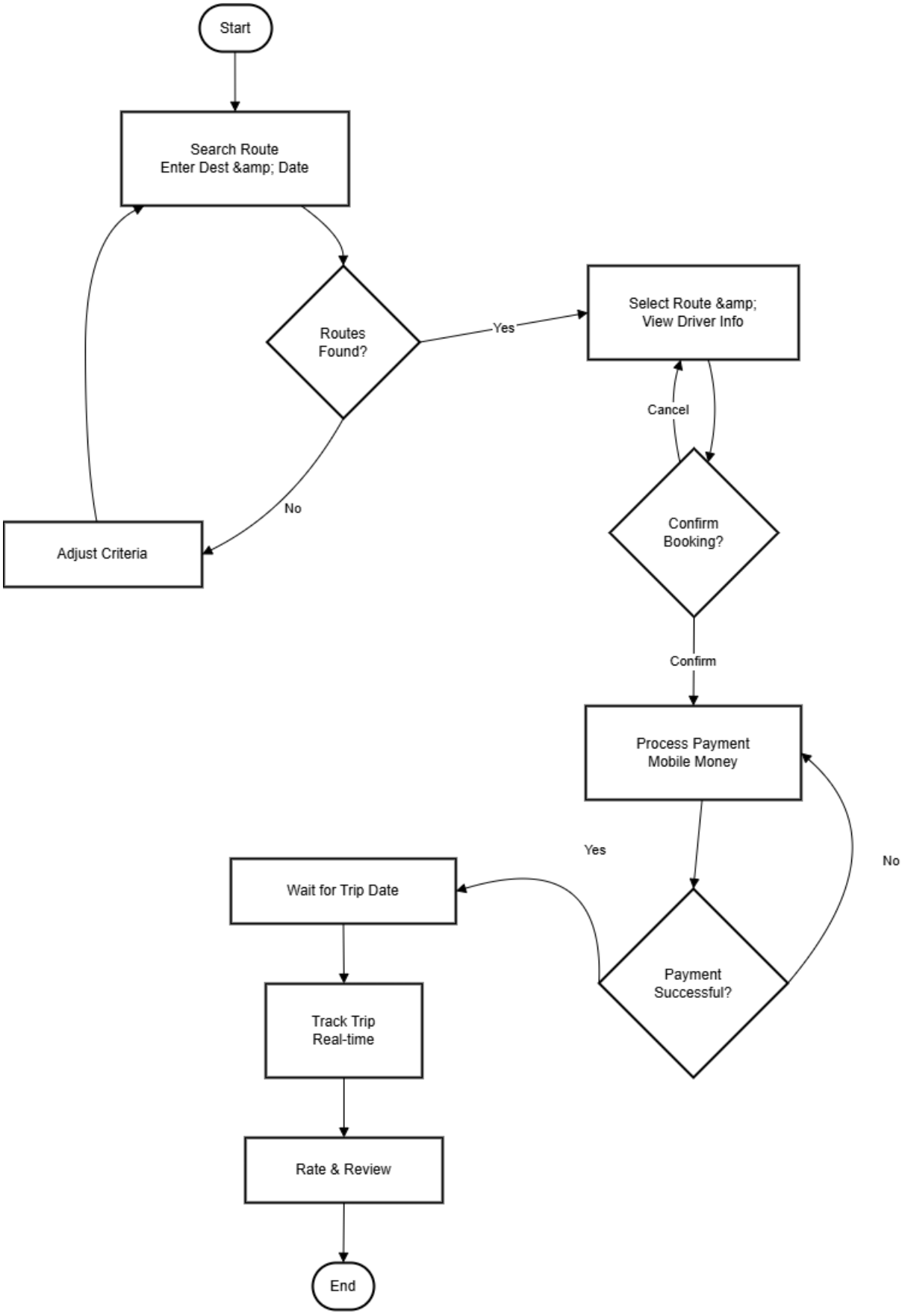


Figure 4.6: Passenger Booking Process Activity Flow

4.3.4 Sequence Diagrams

Sequence diagrams illustrate the interaction flow between system components for specific scenarios.

Booking Process Sequence:

The booking sequence begins when the passenger opens the app and initiates login. The Mobile App sends credentials to the Auth Service, which validates them against the Database and returns a JWT token. The app displays login success. The passenger then enters destination and date, triggering a search request from the app to the Route Service. The Route Service forwards the request to the Matching Engine, which queries available routes from the Database. Matched routes are returned through the chain back to the passenger. The passenger selects a route and the app sends a booking request to the Booking Service. The Booking Service checks seat availability in the Database. If seats are available, the Booking Service notifies the driver via the Notification Service. The driver accepts the booking, and the acceptance is communicated back to the Booking Service. The Booking Service then initiates payment through the Payment Service, which creates a payment record and redirects the passenger to complete payment. Upon successful payment, the Payment Service updates the payment status in the Database and notifies the Booking Service. The Booking Service creates the booking record and triggers confirmation notifications to both the passenger and driver through the Notification Service. Finally, booking details are displayed to the passenger in the app.

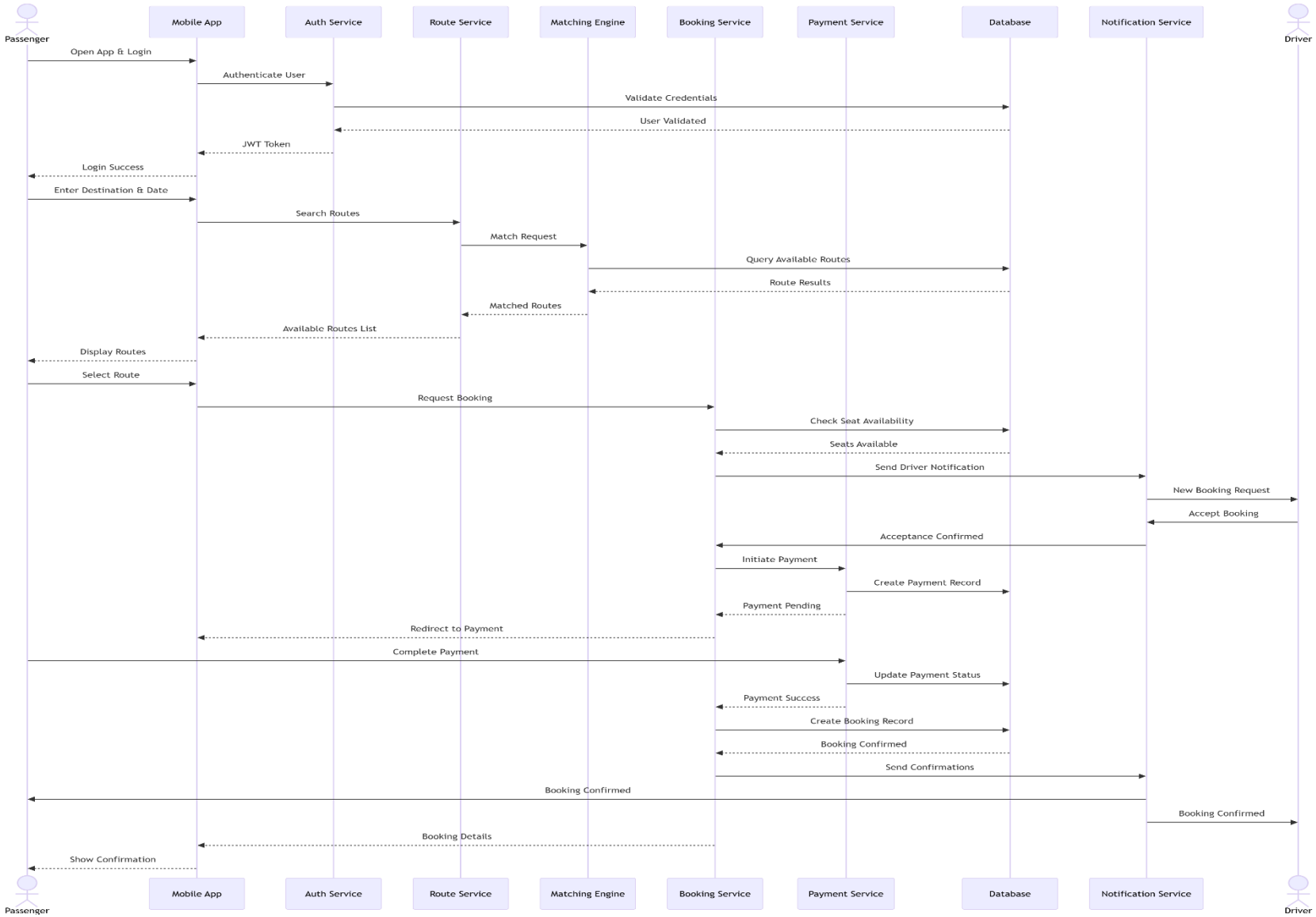


Figure 4.7: Booking Process Sequence Diagram

4.3.5 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) models the database structure, defining entities, attributes, primary keys, foreign keys, and relationships.

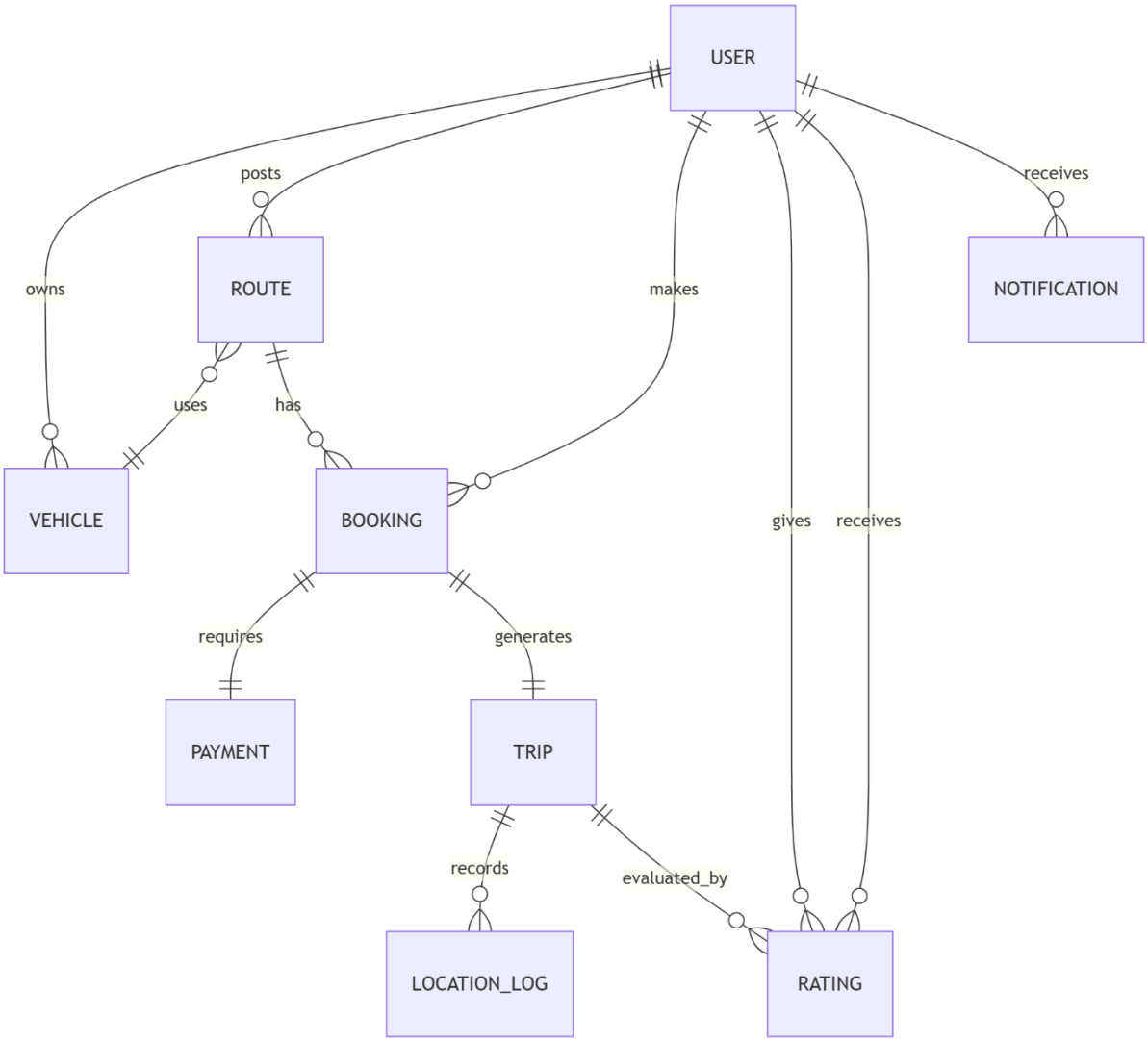


Figure 4.8: Database Entity Relationship Diagram

4.4 Chapter Summary

This chapter presented a comprehensive analysis and design of the Rishfy platform. System analysis identified business requirements aligned with Dar es Salaam's transport crisis, user requirements from survey data, detailed functional requirements across eight modules, and non-functional requirements for performance, security, reliability, usability, scalability, and regulatory compliance. System design then modeled the architecture through use case diagrams, context and Level 1 data flow diagrams, activity diagrams for driver and passenger

workflows, sequence diagrams for booking processes, and a complete entity relationship diagram defining nine core entities and their relationships. Additional design coverage included database decisions, security architecture, and interface design principles. Together, these models provide a complete blueprint for implementing Rishfy as a trust-enabled, LATRA-compliant, and user-centered ride-sharing platform for Tanzania's urban commuters.

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